

YEAMIN HOSSAIN SHIHAB



Dhaka, Bangladesh | +8801521331128 | yeaminionpf.netlify.app | yeaminhs11@gmail.com | in/yeaminhs

Work Experience

Intern at IT Division – PRIME BANK

March 2026 – MAY 2026

- Working on several departments of IT Division such as Project Governance, Enterprise Application Support, Development and Digital Initiatives etc. As I was assigned under Development and Digital Initiatives team, I have build an AI Agent which is capable to do query from PG database by itself. Moreover, I have build an extension for Visual Studio IDE where it will work like cursor/copilot with local LLM. Besides I have build a plugin name OLLAMA-JET which will capable of all understanding the JetBrains IDE and act as an coding assistant.

Undergraduate Teaching Assistant – BRAC University (MNS and CSE)

Jan 2024 – May 2025

- Served as an Undergraduate Teaching Assistant both at Mathematical and Natural Science Department and Computer Science and Engineering Department while supporting instruction, grading, and student engagement for Introduction to Statistics, Complex Variables & Laplace Transformations, Discrete Mathematics and Principles of Physics through tutorials, academic assistance, and collaboration with faculty.

Technical Skills

Programming & Scripting Languages: Python, JavaScript, C, kotlin, dart, R, SQL, Shell Scripting, Assembly (MIPS), Google Apps Script.

Frameworks & Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, FastAPI, Django, Node.js, Express.js, React.js, TensorFlow.js.

Database: MySQL, MongoDB.

Machine Learning & AI: Regression, Classification and Clustering Models. Ensemble Methods (Random Forest, XGBoost, LightGBM) . Deep Learning (CNN, R-CNN, Faster R-CNN, YOLO) . DL Backbones (Inception V3, VGG-16, MobileNet, DenseNet). Model Deployment, Evaluation & Optimization.

LLM & Generative AI Technologies: Flowise, OpenAI API (GPT-4/4o), Hugging Face Transformers, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embedding Models.

Cloud & DevOps: Docker, GitHub Actions, CI/CD Pipelines.

Mathematical & Analytical Tools: Mathematica, Excel Solver, Python Manim, Descriptive & Inferential Statistics, Data Modeling.

Hardware & Embedded Systems (Additional Experience): Arduino, STM32, Raspberry Pi, Circuit Simulation (LTSpice, Quartus).

Projects

Smart Medicine Assistant with AI Insights – Academic Project

Oct 2024 – Dec 2024

- Integrated Gemini 1.5 and a custom NLP model to create an AI chatbot that predicts diseases based on user symptoms, offering intelligent, real-time health insights.
- Enabled home delivery with payment, real-time order tracking, feedback/rating features, and a personalized dashboard experience across all roles.

University Club Monitoring System – Academic Project

Oct 2024 – Dec 2024

- Real-time room availability, conflict-free scheduling, and dynamic booking management — optimized for campus clubs and events.

- Built a personalized dashboard which is clean, user-focused and providing instant access to bookings, events, and actionable updates for every user role.

A simple CGPA Calculator with Django framework

- Developed a web application using Django, Tailwind CSS, and JavaScript, where students can input completed courses and grades to dynamically calculate and display their CGPA with an interactive UI.

Stress Level Classification Using Machine Learning – Personal Project

Dec 2023

- Built a supervised ML system to predict human stress levels (Normal, Slightly High, High) from physiological and behavioral data.
- Trained and evaluated models including **Logistic Regression**, **K-Nearest Neighbors**, **Naive Bayes**, and **Random Forest**, with Random Forest achieving highest accuracy.

Academic Credentials

BRAC University, Dhaka, Bangladesh

BSc in Computer Science & Engineering

CGPA: 3.86 / 4.00

Feb 2021 – Nov 2025

Metropolitan School & College, Dhaka, Bangladesh

Higher Secondary Certificate

GPA: 4.67 / 5.00

Mar 2018 – April 2020

Shyampur GOVT Model School & College, Dhaka, Bangladesh

Secondary School Certificate

GPA: 4.56 / 5.00

Mar 2018 – April 2020

Thesis and Publication

Leveraging Deep Learning Techniques for Pothole Detection

Supervisor: Dewan Ziaul Karim

Feb 2024 – Feb 2025

- We developed an AI-powered system for automatic pothole detection using deep learning and computer vision. Leveraging various model like YOLOv5,v8,v12,RCNN,and our custom CNN model where our model achieved high accuracy in identifying road surface damage from real-world images.

T. Rahman, Y. H. Shihab, T. Bushra, M. T. Shams and D. Z. Karim, "Lightweight Deep Learning Framework for Pothole Detection and Classification Using CNNs and YOLO Models," 2025 28th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2025, pp. 4214-4219

DOI: 10.1109/ICCIT68739.2025.11491787.

Extracurricular Activities

- Director, Human Resources, BRAC University Computer Club
- Member, Organizing Committee, INTRAHACTIVE1.0
- Senior Executive, Robotics Club of BRAC University

Oct 2023 – Dec 2024

Nov 2024

feb 2022– Feb 2023

Referees

Dewan Ziaul Karim

Senior Lecturer

Department of Computer Science and Engineering

BRAC University

Email: ziaul.karim@bracu.ac.bd

Afia Mubassira Islam

Graduate Research Associate

Department of Electrical & Computer Engineering

The Ohio State University, USA

Lecturer (Study leave)

Department of Computer Science & Engineering

BRAC University

Email: afia.islam@bracu.ac.bd